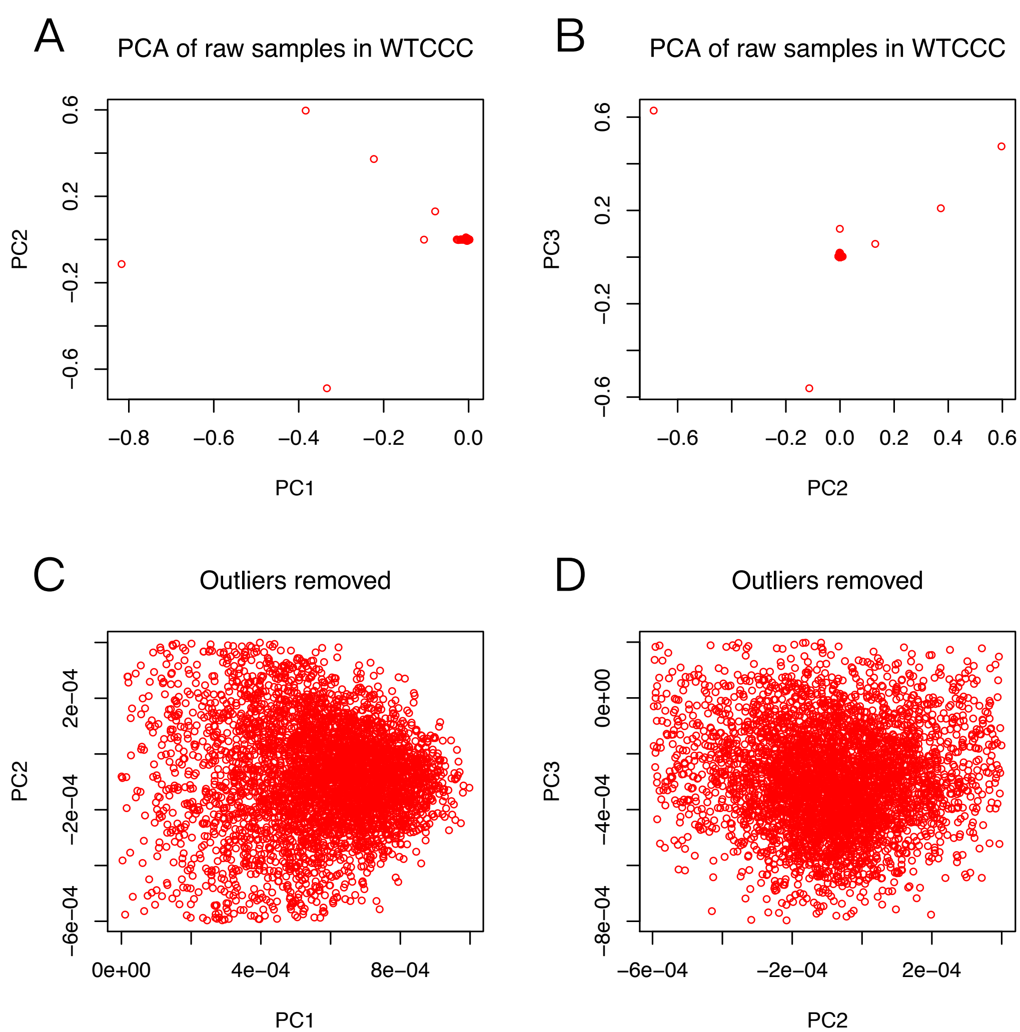
Protocol

1. A British case and control dataset was downloaded from the Wellcome Trust Case Control Consortium (WTCCC) (https://www.wtccc.org.uk) and included 1,999 T2D cases and 3,004 controls scanned for ~500K SNPs.

2. We performed principal components analysis (PCA) using GCTA to remove outliers. Principal component (PC) values of included subjects: (1) 0 ≤ PC1 ≤ 0.001, (2) -0.0006 ≤ PC2 ≤ 0.0004, (3) -0.0008 ≤ PC2 ≤ 0.0002. After filtering out outliers, ~1,600 cases and ~2,500 controls were retained (shown in S1 Figure).

(1) S1 Figure



(2) Command (LINUX - X86_64 X86_64)

Platform: SKLMG10 3.10.0-514.10.2.el7.x86_64 #1 SMP Fri Mar 3 00:04:05 UTC 2017 x86_64 GNU/Linux

Command line:

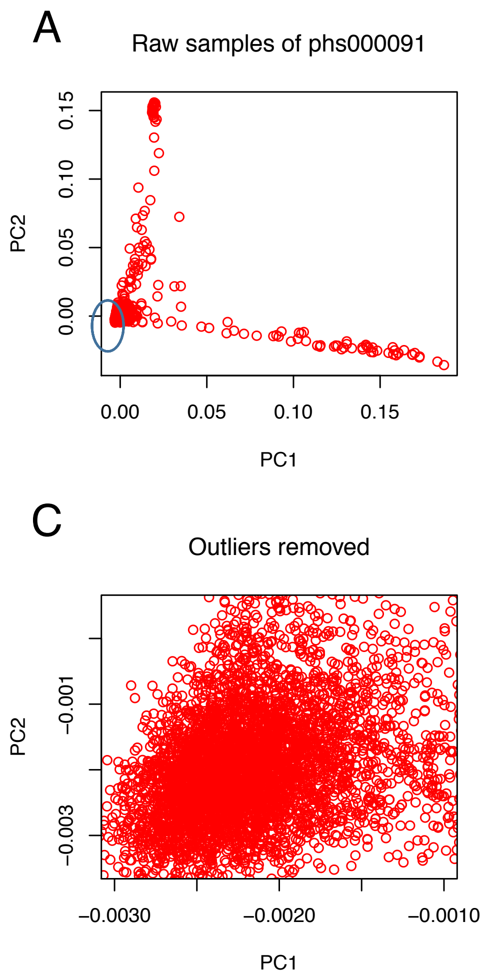
```
/opt/tools/seq-analysis/gcta64 --bfile WTCCC --make-grm --out WTCCC --thread-num 8 --autosome; wait;
/opt/tools/seq-analysis/gcta64 --grm WTCCC --pca 3 --out WTCCC --thread-num 8
```

3. They were then separated into two equal size subgroups at random: one for training and the other for validation. Training cohort consisted of ~800 cases and ~1,300 controls and validation cohort ~800 cases and ~1,300 controls. Training cohort and validation cohort shared no overlapped samples.

4. Another independent dataset of T2D case and control cohorts was also downloaded from dbGaP (https://www.ncbi.nlm.nih.gov/gap). As for phs000091, the actual samples available for downloaded were 2,680 cases and 3,148 controls.

5. Principal component values of included subjects: (1) $-0.003 \leq PC1 \leq -0.001$, (2) $-0.0035 \leq PC2 \leq 0.0005$, (3) $0 \leq PC3 \leq 0.01$. After filtering out outliers by PCA analysis, ~1,700 cases and ~2,000 controls were retained (shown in S2 Figure). The WTCCC dataset and phs000091 dataset shared ~450,000 SNPs. The final two datasets used are shown in Table 1.

(1) S2 Figure



(2) Table 1

Table 1. Basic characteristics of samples used in the study.

	Training	Validation	
	WTCCC	**WTCCC**	**phs000091**
Cases	829	820	1,707
Controls	1,270	1,279	2,042
SNPs	411,165	411,165	703,407

6. Quality control. PLINK was used to remove SNPs in Hardy-Weinberg disequilibrium (Chi-square test P-value < 0.0001 in cases or controls), with > 5% missing data, or with minor allele frequency (MAF) > 0.01. Only autosome SNPs were used. Samples with > 10% missing SNPs and non-founders were excluded (i.e., only parents were retained in cases where their children were also sampled). The cleaned datasets were detailed in Table 1.

(1) Command (MAC OS - MACBOOK PRO (RETINA, 15-INCH, EA))

Platform: Darwin bogon 16.7.0 Darwin Kernel Version 16.7.0: Thu Jun 15 17:36:27 PDT 2017; root:xnu-3789.70.16~2/RELEASE_X86_64 x86_64

The script:

```
#!/bin/bash
#program:
#	data clean of snp of plink bfile
#$1=phsXX   # e.g. phs91 which include phs91.bed,phs91.bim and phs91.fam
###step1 Hard-Weinberg Equibilian####
plink --bfile $1 --hwe 0.0001 --make-bed --out $1-hwe
###step2 remove individual with too much missing genotype data###
plink --bfile $1-hwe --mind 0.1 --make-bed --out $1-hwe-mind
#rm $1-hwe.bed $1-hwe.bim $1-hwe.fam
###step3 exclude snps on their missing genotype rate####
plink --bfile $1-hwe-mind --geno 0.05 --make-bed --out $1-hwe-mind-geno
#rm $1-hwe-mind.bed $1-hwe-mind.bim $1-hwe-mind.fam
###step4 exclude MA of MAF less than 10E-6#####
grep '1$' $1-hwe-mind-geno.fam > control-group.fam.txt
plink --bfile $1-hwe-mind-geno --keep control-group.fam.txt --make-bed --out 
control-group
plink --bfile control-group --freq --out control-group
#awk '$5>0.000001' control-group.frq > control-group.frq.snp.txt
awk '$5>0.01' control-group.frq > control-group.frq.snp.txt
####step5 only snps in chr1-chr22 to be included#####
plink --bfile $1 --extract control-group.frq.snp.txt --make-bed --out $1-hwe-mind-
geno-
maf-control
awk '$1>0 && $1<23' $1-hwe-mind-geno-maf-control.bim > $1-hwe-mind-geno-
maf-control-
autosomal.snp.txt
plink --bfile $1 --extract $1-hwe-mind-geno-maf-control-autosomal.snp.txt --
make-bed --
out $1-hwe-mind-geno-maf-control-autosomal-snp
```

7. MAF refers to the frequency at which the second most common allele occurs in a given population. MA was defined as an allele with MAF < 0.5 in a control group. MAC of an individual was calculated by dividing the number of MAs by the total number of SNPs examined. For calculating mean MAC differences between cases and controls in WTCCC cohorts, the training dataset was merged with the validation dataset. Mean MAC values were compared by *t* test. A two-tailed *P*-value less than 0.05 was considered to indicate statistical significance.

(1) Command (MAC OS - MACBOOK PRO (RETINA, 15-INCH, EA))

Platform: Darwin bogon 16.7.0 Darwin Kernel Version 16.7.0: Thu Jun 15 17:36:27 PDT 2017; root:xnu-3789.70.16~2/RELEASE_X86_64 x86_64

The script:

```
#!/bin/bash
#program:
#	caculate mac based on bfile and maf and deal with the results
#history:
#	2016/3/16 lxy First release
#	2016/5/17 lxy Second release
#use: ./mac.sh data maf
data=$1
dir=$2
mafSmall=$3 # big
mafLarge=$4 # small
#plink=/zs32/home/xylei/leixy/softwares/plink/plink
plink=/Users/leixiaoyun/Downloads/plink/plink
#################caculate MA #########
grep '1$' $dir/${data}.fam > ${data}.control-group.fam.txt
#awk '$3=="half1"' $dir/${data}.fam > ${data}.control-group.fam.txt
$plink --bfile $dir/${data} --keep ${data}.control-group.fam.txt --make-bed --
out ${data}.control-group
#gsed -i 's/half1/0/g'   ${data}.control-group.fam
$plink --bfile ${data}.control-group --freq --out ${data}.control-group
```

```
awk '$5<'"${mafLarge}"' && $5>'"${mafSmall}"' && $1<23 {print $2"\t"$3" "$3}'
${data}.cont
rol-group.frq > ${mafSmall}maf.${mafLarge}maf.snp.ma.txt
##### insert MA and make tfile and last file ########
$plink --bfile $dir/${data}   --extract ${mafSmall}maf.${mafLarge}maf.snp.ma.txt -
-
recode transpose tab --out ${mafSmall}maf.${mafLarge}maf
awk 'BEGIN{FS=OFS="\t"} NR==FNR{a[$1]=$2} NR!=FNR{k=$2;if(k in
a);$4=$4"\t"a[k];print}   '
${mafSmall}maf.${mafLarge}maf.snp.ma.txt
${mafSmall}maf.${mafLarge}maf.tped > insert.tped
cat tfam.header.txt ${mafSmall}maf.${mafLarge}maf.tfam >insert.tfam
/Users/leixiaoyun/Downloads/dist-onlyMA/dist_onlyMA   insert   insert.results
awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[
j,i]" "}print ""}}' ${mafSmall}maf.${mafLarge}maf.tfam >
${mafSmall}maf.${mafLarge}maf.tfa
m.transpos
head -n 6 insert.results| tail -
n 4 | awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){prin
tf a[j,i]" "}print ""}}' |awk '$5=($4-
$3){print}' |awk '$6=($5/$4){print}' > insert.results.transpos.mac
awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[
j,i]" "}print ""}}' insert.results.transpos.mac > insert.results.transpos.mac.transpos
cat ${mafSmall}maf.${mafLarge}maf.tfam.transpos
insert.results.transpos.mac.transpos > ${m
afSmall}maf.${mafLarge}maf.genetic_distance.mac.txt
awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[
j,i]" "}print ""}}' ${mafSmall}maf.${mafLarge}maf.genetic_distance.mac.txt >
${mafSmall}ma
f.to.${mafLarge}maf.genetic_distance.mac.transpos.txt
############ rm #################
rm ${data}.control-group.fam.txt
rm ${data}.control-group.bed ${data}.control-group.bim ${data}.control-group.fam
```

```
rm ${data}.control-group.frq
rm ${mafSmall}maf.${mafLarge}maf.tped ${mafSmall}maf.${mafLarge}maf.tfam
rm insert.tped insert.tfam
rm insert.results ${mafSmall}maf.${mafLarge}maf.tfam.transpos
insert.results.transpos.mac
insert.results.transpos.mac.transpos
#rm    ${mafLarge}maf.tfam.transpos insert.results.transpos.mac
insert.results.transpos.mac.
transpos
rm ${mafSmall}maf.${mafLarge}maf.genetic_distance.mac.txt
rm *.log
```

(2) A custom script was used to calculate the MAC values of each sample(https://github.com/health1987/dist).

8. Linkage disequilibrium (LD) was performed using PLINK for each pair of SNPs in a window of 200kb SNPs; one SNP from the pair was excluded at random if $r^2 > 0.4$. To justify this r2 threshold, we also tested the results at other r^2 levels (i.e. r^2=0.05, r^2=0.2, r^2=0.6 and r^2=0.8).

9. For phs000091, we could download some phenotypic information including age, BMI, alcohol intake, family history of T2D and so on. So for this dataset, we further used multivariate logistic regression test to investigate MAC's role in T2D relative to other risk factors based on R “glm” function.

10. SNPs sets at different *P*-values in training dataset were chosen at first among all SNPs studied here. In addition, LD clumping was performed in WTCCC training cohort. Each MA was given a weighted risk score using the beta value from logistic regression test in PLINK, as described previously. Note that in this case, the MA status was determined using the combined cohort of both cases and controls in the training dataset.

11. Asymptotic *P*-value for each SNP was obtained and different sets of SNPs were chosen to create the genetic risk score at different *P*-value thresholds of <1E-33, <1E-29, <1E-27, <1E-25, <1E-24, <1E-22, <1E-21, <1E-20, <1E-19, <1E-18, <1E-17, <1E-16, <1E-15, <1E-14, <1E-13, <1E-12, <1E-11, <1E-10, <1E-09, <1E-8, <1E-7, <1E-6, <1E-5, <1E-4, <1E-3, <0.01, <0.03, <0.05, <0.07, <0.09, <0.1, <0.3, <0.5, <0.7, <1 and different r2 levels (r2=0.05, r2=0.2, r2=0.4, r2=0.6, r2=0.8). A custom script was used to calculate the total weighted genetic risk score by summing up the beta of each MA.

(1) The formula for calculating genetic risk score is the following:

$$\text{Genetic Risk Score(GRS)} = \sum_{i=1}^{n} \text{beta}_{\text{SNPi}} + 0.5 * \sum_{j=1}^{m} \text{beta}_{\text{SNPj}}$$

SNPi represents MAs in homozygous state and SNPj represents MAs in heterozygous state.

(2) Command (MAC OS - MACBOOK PRO (RETINA, 15-INCH, EA))

Platform: Darwin bogon 16.7.0 Darwin Kernel Version 16.7.0: Thu Jun 15 17:36:27 PDT 2017; root:xnu-3789.70.16~2/RELEASE_X86_64 x86_64

```
#!/usr/bin/env bash
#program:
#       caculate the risk score of specific dataset for cross validation
#history:
#       2016/3/18 lxy First release
#       2016/3/24 lxy Second release
#       2016/3/25 lxy Third release
#       2016/5/20 lxy Fourth release
#       2016/5/20 lxy Fifth release #
data1=$1 # training data i.e.phs92   which includes phs92.bed phs92.bim
phs92.fam
data2=$2      # validation   data   remove   .fam.txt   i.e. phs336
```

```
maf=$3 #maf
pvalue=$4     # p-Value of logit assoction
#data1Dir=$5
#data2Dir=$6
############################---I--training data1-----processing---get-MA--
and--beta--make-
predict-model##########
###step1.1 get MA of specified MAF from control group #############
grep '1$' $5/${data1}.fam > ${data1}.control-group.fam.txt
plink --bfile $5/${data1} --keep ${data1}.control-group.fam.txt --make-bed --
out ${data1}.control-group
plink --bfile ${data1}.control-group --freq --out ${data1}.control-group
awk '($5-'"${maf}"') <0 {print $2" "$3$3" "$3$3}' ${data1}.control-
group.frq > ${data1}.${maf}maf.snp.MAMA.MAMA.txt
rm ${data1}.control-group.fam.txt
rm ${data1}.control-group.bed ${data1}.control-group.bim ${data1}.control-
group.fam
rm ${data1}.control-group.frq
###step1.2 get beta of snp for specified p-value based on maf above
###############
plink --bfile $5/${data1} --extract ${data1}.${maf}maf.snp.MAMA.MAMA.txt --
make-bed --
out ${data1}.${maf}maf
plink --bfile ${data1}.${maf}maf --logistic beta --out ${data1}.${maf}maf
grep -v 'NA' ${data1}.${maf}maf.assoc.logistic   |awk ' ($9-
'"${pvalue}"')<0 {print $7" "$2}'   > ${data1}.${maf}maf.p${pvalue}.beta.snp.txt
rm   ${data1}.${maf}maf.bed ${data1}.${maf}maf.bim ${data1}.${maf}maf.fam
rm ${data1}.${maf}maf.assoc.logistic
###step1.3 merge snp-MAMA-MAMA.file of specified MAF and beta-
snp.file of specified p and MAF #######
awk 'NR==FNR{a[$1]=$2" "$3}NR!=FNR{k=$2;if (k in a);$2=$2" "a[k];print}'
${data1}.${maf}ma
f.snp.MAMA.MAMA.txt ${data1}.${maf}maf.p${pvalue}.beta.snp.txt >
${maf}maf.p${pvalue}.bet
a.snp.MAMA.MAMA.txt
cat   beta.header ${maf}maf.p${pvalue}.beta.snp.MAMA.MAMA.txt   >
${maf}maf.${pvalue}.beta.s
np.ma.ma.txt
rm ${data1}.${maf}maf.p${pvalue}.beta.snp.txt
rm ${maf}maf.p${pvalue}.beta.snp.MAMA.MAMA.txt
```

```
########################--II---validation data2 -----processing---
################
plink --bfile $6/${data2}   --extract   ${maf}maf.${pvalue}.beta.snp.ma.ma.txt    --
recode transpose tab --out validation
cut -f2 validation.tfam > header.1
cat fam.header   header.1   | paste -s - >   header
cat header     validation.tped | sed 's/ //g' |cut -f2,5- > prelast
awk   'NR==FNR{a[$2]=$1" "$2" "$3" "$4} NR!=FNR{k=$1;if(k in a);$1=a[k];print}
' ${maf}maf.
${pvalue}.beta.snp.ma.ma.txt     prelast > last
~/Documents/journal.pone.0133421.s002 last >
${maf}maf.${pvalue}.validation.results
rm   validation.tped
rm header.1
rm prelast #last
rm header
rm ${maf}maf.${pvalue}.beta.snp.ma.ma.txt
###################--III------risk Score results
processing#################
awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[
j,i]" "}print ""}}'   ${maf}maf.${pvalue}.validation.results | awk '$5=($2+$4){print}' |
aw
k
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[j,
i]" "}print ""}}' >   validation.results.transpos
awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[
j,i]" "}print ""}}'   ${maf}maf.${pvalue}.validation.results | awk
'{for(i=1;i<=NF;i++){a[F
NR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){printf a[j,i]" "}print ""}}' >
valid
ation.results.transpos2
cat fam.header2   validation.tfam > validation.add.tfam
awk
'{for(i=1;i<=NF;i++){a[FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){print
f a[
```

```
j,i]" "}print ""}}'    validation.add.tfam > validation.add.fam.transpos
cat validation.add.fam.transpos validation.results.transpos | awk
'{for(i=1;i<=NF;i++){a[F
NR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){printf a[j,i]" "}print ""}}' >
${maf}
maf.${pvalue}.validation.Risk.Score.txt
cat validation.add.fam.transpos validation.results.transpos2 | awk
'{for(i=1;i<=NF;i++){a[
FNR,i]=$i}}END{for(i=1;i<=NF;i++){for(j=1;j<=FNR;j++){printf a[j,i]" "}print ""}}' >
${maf
}maf.${pvalue}.validation.Risk.Score.txt2
awk '{print $6" "$11}' ${maf}maf.${pvalue}.validation.Risk.Score.txt >
${maf}maf.${pvalue}
.validation.Risk.Score.brief.txt
gsed -i '1d' ${maf}maf.${pvalue}.validation.Risk.Score.brief.txt
rm    validation.results.transpos validation.tfam validation.add.tfam
validation.add.fam.tra
nspos
rm *.log #*.hh
rm validation.results.transpos2
```

(3) Command (MAC OS - MACBOOK PRO (RETINA, 15-INCH, EA))

Platform: Darwin bogon 16.7.0 Darwin Kernel Version 16.7.0: Thu Jun 15 17:36:27 PDT 2017; root:xnu-3789.70.16~2/RELEASE_X86_64 x86_64

the perl script:

```
#! /usr/bin/perl
open (INPUT,"@ARGV[0]") or die;
@header;
@outPara1;
@outPara2;
@outPara3;
@outPara4;

sub value {
```

```
    @outPara1;
    @outPara = (@outPara1,@outPara2);
   for($i = 3; $i != $length; ++$i){
        if(@line[$i] =~ /$_[0]/){
            @outPara1[$i] += 1;
        }else {
            @outPara1[$i] += 0;
        }
   }
   return @outPara1;

}

sub weight {
    @outPara2;
   for($i = 3; $i != $length; ++$i){
        if(@line[$i] =~ /$_[0]/){
            @outPara2[$i] +=   @line[0];
        }else {
            @outPara2[$i] += 0;
        }
   }
   return @outPara2;
}

sub valueHet {
    @outPara3;
    @outPara = (@outPara3,@outPara4);
   for($i = 3; $i != $length; ++$i){
        if((@line[$i] =~ /$_[0]/)|(@line[$i] =~ /$_[1]/)){
            @outPara3[$i] += 1;
        }else {
            @outPara3[$i] += 0;
        }
   }
   return @outPara3;

}

sub weightHet {
```

```
    @outPara4;
    for($i = 3; $i != $length; ++$i){
        if((@line[$i] =~ /$_[0]/)|(@line[$i] =~ /$_[0]/)){
            @outPara4[$i] +=    @line[0]/2;
        }else {
            @outPara4[$i] += 0;
        }
    }
    return @outPara4;
}


while(<INPUT>){
    if(grep /rsid/,@line){
        @header = @line;
    }
    @line=split(/ |    /,$_);
    @subline=split(/\t|\n/,$_);
    $length = @line;
    $num_A=grep /A/,@line[3..$length];
    $num_T=grep /T/,@line[3..$length];
    $num_C=grep /C/,@line[3..$length];
    $num_G=grep /G/,@line[3..$length];
    %hash=(
        A=>$num_A,
        T=>$num_T,
        C=>$num_C,
        G=>$num_G,
        );
    my @keys = sort{$hash{$a}<=>$hash{$b}}keys%hash;
    $oldJA=@keys[3];
    $oldMA=@keys[2];

    if(@line[2] =~ /$oldMA$oldMA/){
        $newMA = $oldMA;
        $newJA = $oldJA;
    }else{
        $newMA = $oldJA;
        $newJA = $oldMA;
    }
```

```
    $homoMAC = "$newMA$newMA";
    $het1 = "$newMA$newJA";
    $het2 = "$newJA$newMA";

     @valueHomoMAC = &value($homoMAC);
     @weightHomoMAC = &weight($homoMAC);
     @valueHet1 = &valueHet($het1,$het2);
     @weightHet1 = &weightHet($het1,$het2);

}

for (my $var = 3; $var < $length; $var++) {
    print "\t",@header[$var];
}
print "\nhomoMAC_count\t";
for (my $var = 3; $var < $length; $var++) {
    print @valueHomoMAC[$var],"\t";
}
print "\nhomoMAC_weight\t";
for (my $var = 3; $var < $length; $var++) {
    print @weightHomoMAC[$var],"\t";
}
print "\n";
print "\nhet_count\t";
for (my $var = 3; $var < $length; $var++) {
    print @valueHet1[$var],"\t";
}
print "\nhet_weight\t";
for (my $var = 3; $var < $length; $var++) {
    print $weightHet1[$var],"\t";
}
print "\n";
```

12. Two similar but distinct approaches were performed to estimate the predictive power of the prediction models using the British individuals. For the external cross validation, each model's predictive power was evaluated using the receiver operating characteristic (ROC) curve. Then AUC and the TPR were calculated using the “pROC” R package and Prism 6 (Graphpad).

13. Based on different P-values in the training cohorts of British samples, 210 (35X6=210) models were constructed. AUC and TPR can be obtained for each model in the validation cohort of British samples.

14. In internal 5-fold cross-validation analysis, the training cohort was randomly partitioned into 5 subgroups. Of these, a single subgroup was retained as the validation data for testing the model, and the remaining 4 subgroups were used as training data. Then, the cross-validation process was repeated 5 times, with each of the K subgroups used exactly once as the validation data. The 5 results were averaged to produce a single estimation.

15. The model (i.e. MA set) performing the best in both external cross validation and internal cross validation was chosen as the final risk prediction model.

16. We also compared the prediction accuracy with other PRS based methods (such as PRSice). In addition, for the best risk model, we also used Nagelkerke R2 to evaluate its performance base on “fmsb” R package, which denotes the variance explained in disease state by the GRS or PRS.

(1) Command (MAC OS - MACBOOK PRO (RETINA, 15-INCH, EA))

Platform: Darwin bogon 16.7.0 Darwin Kernel Version 16.7.0: Thu Jun 15 17:36:27 PDT 2017; root:xnu-3789.70.16~2/RELEASE_X86_64 x86_64

one example of using the PRSice software

```
R --file=/Users/leixiaoyun/Downloads/PRSice_v1.25/PRSice_v1.25.R -q
```

```
--args plink /Users/leixiaoyun/Downloads/PRSice_v1.25/plink_1.9_mac_160914
base base_gwas.assoc target ../../validation1 slower 0 sinc 0.001 supper
0.00101    report.individual.scoresT report.best.score F clump.snps F
```

17. In addition, for the model performing the best in the British populations, its predictive power was also estimated in the other one independent cohort phs000091.